

Augmented Auditory Feedback Towards Ball Sports Visual Skill Training

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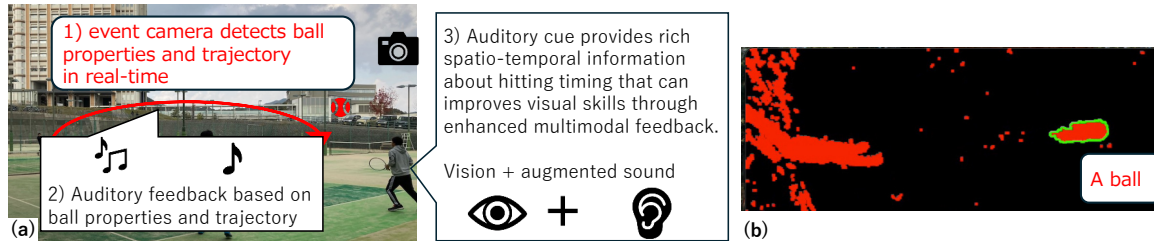


Figure 1: (a) System concept showcasing real-time auditory feedback to enhance visual timing skills using event cameras. (b) Prototype implementation showcasing ball detection during a throw.

ABSTRACT

This position paper introduces the concept of a real-time auditory feedback system designed to promote physical activity and improve well-being through visual skill development in ball sports. The system uses event cameras to translate ball trajectories into intuitive sound cues and complements visual perception with high-temporal-resolution auditory feedback, enabling precise timing and enhanced engagement during practice. Unlike conventional setups requiring wearables or complex hardware, this approach ensures seamless interaction while preserving natural movement. This multimodal approach can potentially offer a scalable and immersive solution for various sports and real-world environments.

Index Terms: Seamless interaction, Auditory feedback, Event camera, Visual skill training.

1 INTRODUCTION

Modern lifestyles often lead to insufficient physical activity due to technological advances. Insufficient physical activity is associated with an increased risk of non-communicable diseases and mental and health problems, according to Strain et al. [18]. They point out that nearly one-third of adults worldwide do not meet recommended physical activity levels, underscoring a critical public health challenge. They also warn that the global estimate of the cost of physical inactivity to public health systems between 2020 and 2030 is approximately US\$27 billion per year if the state of physical inactivity is not changed. Therefore, physical inactivity is an important issue to address.

Playing sports is a good option to be physically active and enjoy various benefits, such as enhanced cardiovascular health and mental well-being [1]. Also, sports provide psychological and social benefits [3], which can help maintain motivation and engagement in long-term participation. Specifically, ball sports appeal because of their diverse nature: they combine physical exertion, strategic elements, and social dynamics through teamwork or rivalry compared to individual activities such as running.

However, ball sports present unique perceptual-cognitive challenges that significantly influence both skill acquisition and par-

ticipation levels. Successfully manipulating a ball requires not only complex motor coordination but also advanced perceptual-cognitive processing under strict temporal constraints [11]. While developing the ability to anticipate and time actions accurately is critical for ball sports, acquiring this temporal-predictive skill poses particular challenges. This is primarily because it depends on both extensive practice and the capacity to process rapidly changing environmental cues. By addressing these perceptual-cognitive challenges effectively, we can potentially facilitate skill development, promote sustained engagement in ball sports, and ultimately contribute to individuals' overall well-being.

To this end, we propose the concept of real-time auditory feedback of ball trajectory using an event camera (see Fig.1). The system augments visual perception with temporal information by converting high-resolution ball trajectory data into auditory cues. This multimodal feedback helps players refine their timing and spatial awareness, improving overall performance and engagement and developing visual skills. While vision plays a fundamental role in ball tracking and movement timing [5], the superior temporal resolution of the auditory system [17] can provide richer information about the temporal dynamics of ball movement. Auditory information is better suited to guide temporally precise behaviors such as sensorimotor synchronization [2], and research suggests that auditory feedback can enhance visual performance by supporting visual timing perception [16]. Based on these findings, we hypothesize that multimodal feedback using the temporal precision of auditory cues can enhance timing based on visuospatial perception. By integrating auditory and visual information, the modalities can complement each other to improve precision in timing tasks. This strategy may improve sensorimotor synchronization and performance in tasks requiring precise temporal-spatial coordination, such as ball sports.

To fully leverage the high temporal resolution of the auditory system, a high-speed sensing system capable of microsecond temporal precision is essential. Event cameras, which detect brightness changes asynchronously with microsecond resolution [4], are well suited for this purpose. Transforming ball motion parameters into corresponding audio signals allows for faster and more precise player responses. At the same time, interactive audio elements increase user engagement. This real-time feedback has advantages over traditional training approaches that rely on post-performance video analysis or verbal instruction, as the immediate nature of auditory feedback allows players to make immediate adjustments during practice sessions, potentially accelerating the overall skill acquisition process.

We envision this auditory feedback system being applied to var-

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ious ball sports, including tennis, badminton, baseball, basketball, and more. In racquet sports, the system aims to improve players' timing and rhythm by augmenting their visual perception with real-time sound. For example, by translating the trajectory of overhead shots into audible cues, players can better judge the ball's descent and refine their stroke technique. In addition, in basketball, providing auditory feedback on the ball's motion during free throws could facilitate the development of embodied knowledge - where players develop an intuitive sense of the ball as an extension of their body. The system could improve players' muscle memory and overall consistency by providing immediate auditory cues related to the ball's arc or release speed. Beyond its training applications, this system can add an interactive, game-like dimension to sports participation. By augmenting traditional gameplay with dynamic sound elements, our approach can create a multi-sensory, immersive experience that engages players and spectators. This enhanced engagement could transform traditional sports into a more interactive and entertaining experience while maintaining their fundamental essence of playing sports.

2 RELATED WORK

Previous research has investigated ball augmentation technologies, focusing on enhancing user engagement and providing dynamic feedback. Optical see-through displays, such as Laplacian Vision, overlay ball trajectories onto users' field of view, enabling better prediction of future positions [9]. The Bouncing Star project integrated wireless modules into balls to enable augmented sports experiences using visual and auditory feedback [10]. Ohta et al. developed trajectory-changeable balls for entertainment, demonstrating how feedback mechanisms enhance user interaction [15]. While these studies highlight the potential of augmentation technologies, their reliance on complex hardware setups limits practical applications in everyday sports.

Research on sound-augmented sports skill development has also been investigated. Ikeda et al. developed a golf training system using audio-visual feedback and motion tracking to project player postures for closed-skill conditions [8], where activities are performed in stable, predictable environments. In virtual reality settings, studies have examined open-skill sports, which require adaptation to dynamic conditions. Masai et al. investigated the effects of sonification in tennis serve return training in VR, demonstrating its effectiveness in improving hip movement timing for slow serve returns [12]. These systems present implementation challenges in real-world sports practice due to their technical requirements and controlled environments.

To harness human auditory perception for immediate feedback, it is critical to track a ball's trajectory with high temporal resolution and minimal latency. Standard RGB cameras, often limited to 120 frames per second, can struggle in fast-paced scenarios, and high-speed cameras tend to generate an overwhelming amount of data. Recent developments, such as TrackNet [7], have shown that high-speed, small objects like tennis balls can be tracked using 30fps broadcast-quality videos by leveraging trajectory patterns across consecutive frames. This approach reduces computational cost while maintaining high precision, but real-time adaptation with minimal latency remains a challenge. In contrast, event cameras operate and detect only changes in brightness (i.e., "events"), enabling high-speed, low-latency detection of moving objects [4]. It can track fast-moving balls in sports like volleyball, where conventional cameras struggle with motion blur [14]. The tracking was not investigated in real-time, but event cameras capture only changes in brightness at the pixel level, reducing data volume to achieve real-time feedback. This capability makes them effective at detecting ball speed, spin, and trajectory with minimal delay, supporting the rapid generation of auditory cues.

Our system concept leverages event cameras positioned in the

environment to generate auditory feedback, eliminating the need for body-mounted sensors. This design improves user comfort and system accessibility by allowing unrestricted movement. The system also provides real-time trajectory information. By bridging the gap between controlled virtual environments and practical sports applications, an event-camera-based system ensures seamless interaction between physical perception and cyber augmentation.

3 PRELIMINARY EXPERIMENT FOR BALL TRACKING

The prototyping implementation of a system using event cameras requires consideration of the balance between processing speed and accuracy. While advanced techniques like motion estimation using contrast maximization [5] or machine learning models trained on large datasets can offer high precision, they are computationally expensive, making them unsuitable for real-time processing. Therefore, simpler algorithms are needed to achieve efficient processing without sacrificing usability.

We conducted a preliminary experiment to test the feasibility of tracking a thrown ball using an event camera. To effectively manage the sparse yet high-temporal-resolution event data, we accumulated events of 30 ms into frames. The ball detection pipeline consists of several sequential processing stages. Initially, a threshold-based filtering step retains only those non-zero pixels whose local neighborhoods exceed a preset activity threshold, effectively removing spurious or isolated events. The filtered frame then undergoes morphological processing, specifically closing and dilation operations, to fill small gaps and further reduce noise. Next, contours are extracted from the processed frame, and their shape metrics—including circularity, aspect ratio, and area—are evaluated to filter out irrelevant objects. Finally, the system performs event-density-based filtering by checking the event density within and around each contour (within an (n) -pixel buffer), discarding any contour that fails to meet the specified threshold.

By combining these steps, the method localizes and tracks the ball under experimental conditions with a simple fast algorithm (see Fig. 1 (b)), although it sometimes misidentifies the ball's shadow.

4 DISCUSSION AND FUTURE DIRECTIONS

For a ball-tracking algorithm, we only tested specific cases under experimental conditions. Further research is needed to automate and optimize its parameters to ensure robust and real-time processing. We also need to test the feasibility in real-world scenarios.

Designing effective auditory feedback and evaluating the system's impact on skill development is critical. One of the simplest approaches to generating sound feedback involves controlling the pitch of sine waves based on the ball's position. This method can convey the ball's location to the user intuitively. Another possibility is audio volume, spatial orientations, or white-noise "wind" effects. Additionally, providing rhythmic cues can be effective in helping players grasp the optimal hitting timing [13]. An additional layer of "juiciness" - a game design concept for enhancing user engagement [6] - can be adapted to our auditory feedback system. By incorporating playful sound effects that highlight successful hits or ball spin characteristics, we can enhance player motivation and practice engagement.

Long-term experiments are essential to evaluate the system's impact on skill acquisition, engagement, and motivation. Metrics such as swing timing accuracy, reaction speed, and player feedback will help establish the system's effectiveness across diverse sports.

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